

# **JMO – 1997**

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1.

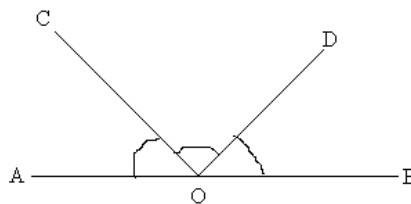
- (a) The cost price of 6 pens and 9 pencils is Rs. 41.25 and that of 8 pens and 5 pencils is Rs. 42.75. Find the price of each of the pen and pencil separately.
- (b) Fill the missing digits denoted by \* in the following division.

$$\begin{array}{r}
 3 * 7 \overline{) \begin{array}{r} * 8 * \\ 5 * 6 * 7 \\ 3 * 7 \\ \hline 2 7 9 * \\ 2 * 3 6 \\ \hline 2 5 * 7 \\ 2 * * 6 \\ \hline 2 1 \end{array}}
 \end{array}$$

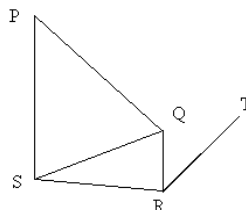
- (c) Plantation of trees is to be made at 3 metres apart in a rectangular field of length 24 metres and breadth 18 metres. If the cost of planting a tree is Rs. 3.75 then find out the total cost of plantation of the field.
- (d) The simple interest on a certain sum for 5 years at 8% per annum is Rs. 200/- less than the simple interest on the same amount (sum) for 3 years and 4 months at 18% per annum. Find out the sum.
- (e) A boy was asked to find out LCM of 8, 12 and third number. While finding out the LCM of these three numbers the boy wrote 21 in place of 12 by mistake. However, he got the correct answer. Find out the smallest value of third number and also find the LCM of three numbers.

2.

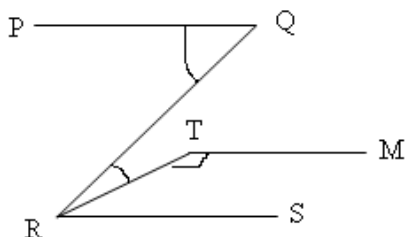
- (a) In the given figure, AOB is a straight line. If  $m\angle AOC + m\angle COD = 100^\circ$ , and  $m\angle BOD + m\angle COD = 146^\circ$ , then find out  $m\angle COD$ .



- (b) In the given figure  $RT \parallel SQ$ . If  $m\angle QPS = 100^\circ$ ,  $m\angle PQS = 40^\circ$ ,  $m\angle PSR = 85^\circ$ ,  $m\angle QRS = 70^\circ$ , then find out  $m\angle QRT$ .



- (c) In the given figure  $PQ \parallel TM \parallel RS$ . If  $m\angle PQR = 70^\circ$ ,  $m\angle RTM = 140^\circ$ , then find out  $m\angle QRT$ .



- (d) A box contains 10 red, 10 green, 6 white balls. What is the smallest number of balls that must be drawn without looking if at least 8 balls are to be of same color?
- 3.
- (a) The total cost price of 2 watches is Rs. 3360/-. By selling one at a loss of 12% and the other at a gain of 16% there is neither loss nor gain in the total. Find out the cost price of each watches.
- (b) If Ram goes from his home to school at a speed of 2 km/h then he is 6 minutes late, but if he goes at a speed of 3 km/h then he reaches the school 10 minutes early. What is the distance between his school from his home?
- (c) Insert (+) or (-) sign between the numbers so that left hand side will be equal to the right hand side number of the equal sign.  $57 \quad 115 \quad 199 \quad 137 \quad 178 \quad = \quad 138$ .
4. A man started from home at 2.35PM (by his clock at home) and drove to his village and arrived there when the village clock indicated 3.10PM. After staying there for 25 minutes he drove back at a speed of twice as fast as that of previous speed, reaching home at 4. 00 PM (by his clock at home). Find out how much the village clock was fast or late in comparison to his clock at home.
- 5.
- (a) In an examination a student appeared five papers such as Mathematics, Science, Social Studies, History and Sanskrit. Where the full marks were same for each papers. His marks obtained in these papers were in proportion 6:7:8:9:10. He obtained  $\frac{3}{5}$  th part of the total marks. Find out the papers where he got more than 60% of marks.
- (b) In an election 10% of the voters on the voter list did not cast their votes and 60 votes were rejected. There were only two candidates. The winner was supported by 47% of all voters in the list and he got 308 votes more than his rival. Find out the number of voters in the list.
6. A farmer engages 4 men and 10 women to gather in his harvest. At the end of 6 days one half of the harvest is gathered in and two more men and two more women were again engaged. In the next 3 days another one third of the harvest is gathered in. How many more women must be engaged if the whole of the harvest must be completed in 1 day more?
7. In the number  $30 * 0 * 03$ , chose the two missing digits \*in such way that the number will be divisible by 13. Find out at least two solutions.
- 8.
- (a) A certain 50 digits number has 1 in 30 digits (places) and 0 in the rest of digits. Can it be perfect square? Justify your answer.
- (b) Find out the four digit number which when divided by 131 and 132 will leave remainder 112 and 98 respectively.

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